

BILKENT UNIVERSITY  
ELECTRICAL AND ELECTRONICS ENGINEERING

**MATH 241**  
**ENGINEERING MATHEMATICS**

FINAL EXAM  
JULY 31, 2014

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**Instructions:**

- There are 5 questions in the exam.
  - Duration of the exam is 160 minutes.
  - The exam is closed book, closed notes.
  - No calculators are allowed.
  - Make sure that your cellphone is turned off before starting the exam.
  - Please read each question carefully and explain your answers clearly in order to get credit.
  - Do not detach pages.
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Surname : ... **SOLUTION MANUAL** .....  
Name : .....  
ID : .....  
Signature : .....

| Question            | Grade |
|---------------------|-------|
| 1 <b>10</b>         |       |
| 2 <b>25</b>         |       |
| 3 <b>25</b>         |       |
| 4 <b>20</b>         |       |
| 5 <b>20</b>         |       |
| <b>Total: (100)</b> |       |

Q1. Let  $A$  be an  $m$  by  $n$  matrix and the rank of  $A$  be  $r$ .

- (a) (2 Points) What are the dimensions of  $\text{null}(A)$ ,  $\text{null}(A^T)$ ,  $\text{row}(A)$  and  $\text{col}(A)$ ?  
 (b) (2 Points) Show that  $\text{null}(A)$  is orthogonal to  $\text{row}(A)$ .  
 (c) (2 Points) Show that  $\text{null}(A^T)$  is orthogonal to  $\text{col}(A)$ .  
 (d) (2 Points) Let  $x$  be any vector in  $\mathbb{R}^n$ ,  $x_p \in \text{row}(A)$  and  $x_h \in \text{null}(A)$ , briefly explain whether  $x$  can be expressed as a linear combination of  $x_p$  and  $x_h$ .  
 (e) (2 Points) Let  $y_p \in \text{col}(A)$ ,  $y_h \in \text{null}(A^T)$  and  $y = \alpha y_p + \beta y_h$ . Find all possible values for  $\alpha$  and  $\beta$  such that there exists a solution to the system of linear equations  $Ax = y$ . In case the solution does not exist, what can be done to find the best approximation for the solution?

(a)  $\dim(\text{Row}(A)) = \dim(\text{Col}(A)) = \# \text{ of LI rows or columns} = r$

$\dim(\text{Null}(A)) = \# \text{ of free variables} = n - r$

$\dim(\text{Null}(A^T)) = \# \text{ of free variables for } A^T = m - r$

(b)  $\left. \begin{array}{l} y \in \text{Null}(A) \Rightarrow Ay = 0 \\ u \in \text{Row}(A) \Rightarrow u = A^T \alpha \end{array} \right\} u \cdot y = u^T y = \alpha^T A y = \alpha^T 0 = 0$       c)  $\left. \begin{array}{l} v \in \text{Null}(A^T) \Rightarrow A^T v = 0 \\ u \in \text{Col}(A) \Rightarrow u = A \alpha \end{array} \right\} u^T v = \alpha^T A v = \alpha^T 0 = 0$

(d) Yes, it can be expressed as a linear combination of  $x_p$  and  $x_h$  since

$\dim(\text{Null}(A) \cup \text{Row}(A)) = n$

(e) Since  $y$  should be in  $\text{Col}(A)$  for a solution,  $\beta$  should be zero while  $|\alpha| \neq \infty$  or

If the solution does not exist, we always project  $y$  to  $\text{Col}(A)$  orthogonally, then solve

for  $Ax = \alpha y_p$ .

Q2. Consider the following matrix and the vector.

$$A = \begin{bmatrix} 1 & -2 & -2 \\ 2 & -1 & 2 \\ 2 & 2 & 8 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}$$

Let the column space of  $A$  and the row space of  $A$  are denoted as  $\text{col}(A)$  and  $\text{row}(A)$  respectively.

(a)

- 2 i. Find a basis for  $\text{null}(A)$ .
- 2 ii. Find a basis for  $\text{row}(A)$ .
- 2 iii. Find a basis for  $\text{null}(A^T)$ .
- 2 iv. Find a basis for  $\text{col}(A)$ .
- 4 v. Verify that  $\text{null}(A)$  and  $\text{null}(A^T)$  are orthogonal to  $\text{row}(A)$  and  $\text{col}(A)$  respectively.

(b) Let  $\mathbf{b}_c$  be the orthogonal projection of  $\mathbf{b}$  onto  $\text{col}(A)$ .

- 4 i. Find  $\mathbf{b}_c$ .
- 2 ii. Verify that  $\mathbf{b}_c$  is orthogonal to  $\text{null}(A^T)$ .
- 5 iii. Find the particular solution for  $A\mathbf{x} = \mathbf{b}_c$  that is in  $\text{row}(A)$ , i.e.,  $A\mathbf{x}_p = \mathbf{b}_c$  where  $\mathbf{x}_p \in \text{row}(A)$ .
- 2 iv. Verify that  $\mathbf{x}_p$  is orthogonal to  $\text{null}(A)$ .

$$(a) \quad A = \begin{bmatrix} 1 & -2 & -2 \\ 2 & -1 & 2 \\ 2 & 2 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -2 \\ 0 & 3 & 6 \\ 0 & 6 & 12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -2 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & N \\ 0 & 0 \end{bmatrix}$$

i.  $\text{Null}(A) = \text{span} \left\{ \begin{bmatrix} -N \\ I \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \right\}$

ii.  $\text{Row}(A) = \text{span} \left\{ \begin{bmatrix} I \\ N^T \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 6 \end{bmatrix} \right\}$

$$A^T = \begin{bmatrix} 1 & 2 & 2 \\ -2 & -1 & 2 \\ -2 & 2 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & 6 \\ 0 & 6 & 12 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & -2 \\ 0 & 3 & 6 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} I & M \\ 0 & 0 \end{bmatrix}$$

(iii)  $\text{Null}(A^T) = \text{span} \left\{ \begin{bmatrix} -M \\ I \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \right\}$

(iv)  $\text{Col}(A^T) = \text{span} \left\{ \begin{bmatrix} I \\ M^T \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix} \right\}$

(b) (i) Let columns of  $C$  form a basis for  $\text{col}(A)$ , the  $C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -2 & 2 \end{bmatrix}$

$$\mathbf{b}_c = C\mathbf{z} \text{ such that } C^T C \mathbf{z} = C^T \mathbf{b} \Rightarrow \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix} \mathbf{z} = \begin{bmatrix} -3 \\ 9 \end{bmatrix} \Rightarrow \mathbf{z} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\mathbf{b}_c = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}$$

$$(ii) \quad \mathbf{v} \in \text{Null}(A^T) \Rightarrow \mathbf{v} = \alpha \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} \quad \mathbf{v}^T \mathbf{b}_c = \alpha \begin{bmatrix} 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix} = \alpha (-2 - 2 + 4) = \alpha \cdot 0 = 0$$

$$(iii) \quad \begin{bmatrix} 1 & -2 & -2 & -1 \\ 2 & -1 & 2 & 1 \\ 2 & 2 & 8 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 3 & 6 & 3 \\ 0 & 6 & 12 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & -2 & -1 \\ 0 & 3 & 6 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 3 & 6 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix}$$

$$\mathbf{x}_n = \left( \frac{1}{9} \begin{bmatrix} -2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} = -\frac{4}{9} \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \text{ projection onto } \text{Null}(A)$$

$$\mathbf{x}_p = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \frac{4}{9} \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} = \mathbf{x}_0 - \mathbf{x}_n \rightarrow \text{subtract null}(A) \text{ component to get Row}(A) \text{ component of } \mathbf{x}_0.$$

$$(iv) \quad \mathbf{v} \in \text{Null}(A) \Rightarrow \mathbf{v} = \alpha \begin{bmatrix} -2 \\ -2 \\ 1 \end{bmatrix} \quad \mathbf{v}^T \mathbf{x}_p = \alpha \begin{bmatrix} -2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} \cdot \frac{1}{9} = \frac{\alpha}{9} (-2 - 2 + 4) = 0$$

Q3. Let  $A$  be an  $n$  by  $n$  matrix,

- (a) (4 points) Show that if  $A$  has  $n$  linearly independent eigenvectors then  $A$  is diagonalizable.  
 (b) (7 points) Prove that if the eigenvalues of  $A$  are all distinct then the eigenvectors of  $A$  are linearly independent.  
 (c) (4 points) If  $A$  has only one eigenvalue with multiplicity  $n$  and the dimension of the eigenspace of  $A$  is  $n$ , then show that  $A = \lambda I$ .  
 (d) (4 points) For a symmetric  $A$  that has distinct eigenvalues, show that the eigenvectors form an orthonormal basis for the eigenspace of  $A$ .  
 (e) (6 points) Let  $A = \begin{bmatrix} -a_1 & -a_2 & \dots & -a_{n-1} & -a_n \\ & I & & & 0 \end{bmatrix}$  where  $I$  is identity matrix of size  $(n-1)$  and  $0$  is the zero vector of size  $n-1$ , then show that the characteristic equation of  $A$  is equal to  $s^n + a_1 s^{n-1} + a_2 s^{n-2} + a_3 s^{n-3} + \dots + a_{n-1} s + a_n = \det(sI - A)$ .

(a)  $A v_i = \lambda_i v_i \quad 1 \leq i \leq n \Rightarrow A [v_1 \ v_2 \ \dots \ v_n] = [v_1 \ v_2 \ \dots \ v_n] \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_n \end{bmatrix} \Rightarrow A V = V D$   
 since  $V$  is invertable (its columns are LI),  $V^{-1} A V = D$  so we can diagonalize  $A$ .

(d)  $v_j^T A v_i = \lambda_i v_j^T v_i = (A^T v_j)^T v_i = (A v_j)^T v_i = \lambda_j v_j^T v_i \Rightarrow (\lambda_i - \lambda_j) v_j^T v_i = 0 \Rightarrow v_j^T v_i = 0$   
 $\downarrow$   
 $A$  symmetric  $\neq 0$  for  $i \neq j$

(c) Since all eigenvectors are LI, so  $A$  can be diagonalizable, and we have only one eigenvalue  $\lambda$ , then  $D = \lambda I$ .  $A = V D V^{-1} = V \lambda I V^{-1} = \lambda V V^{-1} = \lambda I$

(b) Assume they are LD then there exists a set of coefficients  $\alpha_i$  not all equal to zero satisfies  $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$ ,  $E = [\alpha_1 v_1 \ \alpha_2 v_2 \ \dots \ \alpha_n v_n] \Rightarrow E \cdot \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$   $\lambda = \begin{bmatrix} \lambda_1^n \\ \lambda_2^n \\ \vdots \\ \lambda_n^n \end{bmatrix}$   
 Multiply both sides with  $A, A^2, \dots, A^{n-1}$   
 $A E \cdot \underline{1} = E \lambda = 0$   
 $A^2 E \cdot \underline{1} = E \lambda^2 = 0 \dots$  upto  $n$

We have  $E \begin{bmatrix} 1 & \lambda & \lambda^2 & \dots & \lambda^{n-1} \end{bmatrix} = \underline{0}$

Vandermonde matrix and its invertable if  $\lambda_i \neq \lambda_j \quad i \neq j$ , and this is the case here.

So  $E = \underline{0} \rightarrow$  each column of  $E$  is zero  $\Rightarrow \alpha_i v_i = 0$  since  $v_i \neq 0 \quad \alpha_i = 0$  for  $1 \leq i \leq n$   
 Therefore we've reached a contradiction, then the vectors are LI.

(e)  $\left| (sI - A) \right| = \begin{vmatrix} s+a_1 & a_2 & \dots & a_n \\ -1 & s & & \\ & -1 & s & \\ & & \ddots & s \\ & & & -1 & s \end{vmatrix} = (s+a_1) \begin{vmatrix} s & & \\ -1 & s & \\ & -1 & s \\ & & \ddots & s \end{vmatrix} + \begin{vmatrix} a_2 & a_3 & \dots & a_n \\ -1 & s & & \\ & -1 & s & \\ & & \ddots & s \end{vmatrix} = s^n + a_1 s^{n-1} + a_2 \begin{vmatrix} s & & \\ -1 & s & \\ & -1 & s \end{vmatrix} + \begin{vmatrix} a_3 & \dots & a_n \\ -1 & s & \\ & -1 & s \end{vmatrix}$   
 $\begin{matrix} P_2(s) = a_2 s^{n-2} + P_3(s) \\ P_3(s) = a_3 s^{n-3} + P_4(s) \\ \vdots \\ P_{n-1}(s) = a_{n-1} s + a_n \end{matrix} \Rightarrow P_2(s) = a_2 s^{n-2} + a_3 s^{n-3} + \dots + a_{n-1} s + a_n$

Therefore  $\left| sI - A \right| = \sum_{i=0}^n a_i s^{n-i}$  and  $a_0 = 1$ .

Q4.

(a) (6 points) Find the Z transform of  $x[k] = a^k \cos(\theta k)$ .

(b) (6 points) Find the inverse Z transform of  $X(z) = \frac{z^{-1}}{1-z^{-1}+\frac{1}{2}z^{-2}}$ .

(c) Consider the following differential equation,

$$x'' = -x' + 2x$$

i. (6 points) Find the solution for initial conditions  $x(0) = 2, x'(0) = -1$ .

ii. (2 points) Find a suitable non-zero initial condition that makes the system stable, i.e.,  $x(t) \rightarrow 0$  as  $t \rightarrow \infty$ .

$$\begin{aligned} (a) \quad a^k \cos(\theta k) &= a^k \left( \frac{e^{j\theta k} + e^{-j\theta k}}{2} \right) = \frac{1}{2} \left( (ae^{j\theta})^k + (ae^{-j\theta})^k \right) \xrightarrow{Z^{-1}} \frac{1}{2} \left( \frac{1}{1-ae^{j\theta}z^{-1}} + \frac{1}{1-ae^{-j\theta}z^{-1}} \right) \\ &= \frac{1}{2} \frac{2-2a\cos\theta z^{-1}}{1-2a\cos\theta z^{-1}+a^2z^{-2}} = \boxed{\frac{1-a\cos\theta z^{-1}}{1-2a\cos\theta z^{-1}+a^2z^{-2}}} = X(z) \end{aligned}$$

$$\begin{aligned} (b) \quad 1-z^{-1}+\frac{1}{2}z^{-2} &= 0 \quad z = \frac{1 \pm \sqrt{1-2}}{2} = \frac{1 \pm j}{2} = \frac{1}{\sqrt{2}} e^{\pm j\pi/4} \\ \frac{z^{-1}}{1-z^{-1}+\frac{1}{2}z^{-2}} &= \frac{z^{-1}}{\left(1-\frac{1}{\sqrt{2}}e^{j\pi/4}z^{-1}\right)\left(1-\frac{1}{\sqrt{2}}e^{-j\pi/4}z^{-1}\right)} = \frac{A}{\left(1-\frac{1}{\sqrt{2}}e^{j\pi/4}z^{-1}\right)} + \frac{B}{\left(1-\frac{1}{\sqrt{2}}e^{-j\pi/4}z^{-1}\right)} \\ A &= \frac{z^{-1}}{1-\frac{1}{\sqrt{2}}e^{-j\pi/4}z^{-1}} \Bigg|_{z^{-1}=\frac{1}{\sqrt{2}}e^{-j\pi/4}} = \frac{\sqrt{2}e^{-j\pi/4}}{1-e^{-j\pi/2}} = \frac{\sqrt{2}}{\sqrt{2}} \cdot \frac{e^{-j\pi/4}}{e^{j\pi/4}} = \frac{1}{j}, \quad B = A^* = -\frac{1}{j} \end{aligned}$$

$$\mathcal{Z}^{-1}(X(z)) = \frac{1}{j} \left( \left( \frac{1}{\sqrt{2}} e^{j\pi/4} \right)^n - \left( \frac{1}{\sqrt{2}} e^{-j\pi/4} \right)^n \right) = 2 \cdot \left( \frac{1}{\sqrt{2}} \right)^n \sin\left(\frac{\pi}{4}n\right) = \left( \frac{1}{\sqrt{2}} \right)^{n-2} \sin\left(\frac{\pi}{4}n\right) = x[n]$$

$$c) \text{ (i) } x'' = -x' + 2x \Rightarrow s^2 X(s) - s x(0) - x'(0) = -(s X(s) - x(0)) + 2 X(s)$$

$$\Rightarrow (s^2 + s - 2) X(s) = s x(0) + x'(0) + x(0) \quad X(s) = \frac{s x(0) + x'(0) + x(0)}{s^2 + s - 2} = \frac{A}{s+2} + \frac{B}{s-1}$$

$$A = (s+2) X(s) \Big|_{s=-2} = \frac{-2x(0) + x'(0) + x(0)}{-2-1} = \frac{x(0) - x'(0)}{3}; \quad B = \frac{2x(0) + x'(0)}{3}$$

$$x(t) = \frac{x(0) - x'(0)}{3} e^{-2t} + \frac{2x(0) + x'(0)}{3} e^t \quad \xrightarrow{\text{for } x(0)=2, x'(0)=-1} \quad x(t) = e^{-2t} + e^t$$

$$\text{(ii) } \frac{2x(0) + x'(0)}{3} e^t \text{ should be zero} \Rightarrow 2x(0) + x'(0) = 0$$

$$\boxed{\begin{matrix} x'(0) = -2x(0) \\ x(0) = 1 \quad x'(0) = -2 \end{matrix}}$$

$$\boxed{x(t) = e^{-2t}}$$

Q5. Let the system of linear differential equations have the following form,

$$\mathbf{x}' = \mathbf{A}\mathbf{x}$$

where the states vector is defined as  $\mathbf{x} = [x_1(t) \dots x_n(t)]^T$ .

(a) (8 points) Let the eigenvectors of  $\mathbf{A}$  are all linearly independent and the initial condition is expressed as a linear combination of the eigenvectors as,

$$\mathbf{x}(0) = \sum_{i=1}^n \alpha_i \mathbf{v}_i \text{ where } \mathbf{v}_i \text{ is the eigenvector that is associated with the eigenvalue } \lambda_i.$$

Show that the solution to this homogeneous system is given below as,

$$\mathbf{x}(t) = \Phi(t; 0, \mathbf{x}(0)) = \Phi(t; 0)\mathbf{x}(0) = \sum_{i=1}^n \alpha_i e^{\lambda_i t} \mathbf{v}_i$$

where  $\Phi(t; 0) = e^{\mathbf{A}t}$  is the state transition matrix and  $\Phi(t; 0, \mathbf{x}(0))$  is the homogeneous solution with the initial condition  $\mathbf{x}(0)$ .

(b) Let  $\mathbf{A} = \begin{bmatrix} -3 & 4 \\ -1 & 2 \end{bmatrix}$ ,

9 i. Find  $\Phi(t; 0, \begin{bmatrix} -1 \\ 1 \end{bmatrix})$ ,  $\Phi(t; 0, \begin{bmatrix} 1 \\ 1 \end{bmatrix})$  and  $\Phi(t; 0)$ .

3 ii. Find a basis for the initial condition that makes the system stable.

$$(a) \quad \underline{\mathbf{x}}(t) = \underline{\Phi}(t; 0) \underline{\mathbf{x}}_0 = e^{\mathbf{A}t} \cdot \underline{\mathbf{x}}_0 = \underline{\mathbf{V}} e^{\mathbf{D}t} \underline{\mathbf{V}}^{-1} \cdot \underline{\mathbf{V}} \underline{\alpha} = \underline{\mathbf{V}} e^{\mathbf{D}t} \underline{\alpha} = [\underline{\mathbf{v}}_1 \dots \underline{\mathbf{v}}_n] \begin{bmatrix} \alpha_1 e^{\lambda_1 t} \\ \vdots \\ \alpha_n e^{\lambda_n t} \end{bmatrix} = \sum_{i=1}^n \alpha_i e^{\lambda_i t} \underline{\mathbf{v}}_i$$

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2 t^2}{2} + \dots = \mathbf{I} + \mathbf{V} \mathbf{D} \mathbf{V}^{-1} t + \frac{\mathbf{V} \mathbf{D}^2 \mathbf{V}^{-1} t^2}{2} + \dots = \mathbf{V} \left( \mathbf{I} + \mathbf{D}t + \frac{\mathbf{D}^2 t^2}{2!} + \dots \right) \mathbf{V}^{-1} = \underline{\mathbf{V}} e^{\mathbf{D}t} \underline{\mathbf{V}}^{-1}$$

$$e^{\mathbf{D}t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & \ddots \\ & & & e^{\lambda_n t} \end{bmatrix}$$

$$(b) \quad (s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s+3 & -4 \\ 1 & s-2 \end{bmatrix} \Rightarrow |s\mathbf{I} - \mathbf{A}| = (s+3)(s-2) + 4 = s^2 + s - 2 = (s+2)(s-1)$$

$$\text{for } s=1 \quad \begin{bmatrix} 4 & -4 \\ 1 & -1 \end{bmatrix} \underline{\mathbf{v}}_1 = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{v}}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \text{for } s=-2 \quad \begin{bmatrix} 1 & -4 \\ 1 & -4 \end{bmatrix} \underline{\mathbf{v}}_2 = \underline{\mathbf{0}} \Rightarrow \underline{\mathbf{v}}_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$(i) \quad \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix} \underline{\alpha} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \Rightarrow \underline{\alpha} = \frac{1}{3} \begin{bmatrix} -1 & 4 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 5 \\ -2 \end{bmatrix} \Rightarrow \underline{\mathbf{x}}(t) = \underline{\Phi}(t; 0, \begin{bmatrix} -1 \\ 1 \end{bmatrix}) = \frac{5}{3} e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{2}{3} e^{-2t} \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

$$\text{for } \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \alpha_1 = 1, \alpha_2 = 0 \quad \underline{\Phi}(t; 0, \begin{bmatrix} 1 \\ 1 \end{bmatrix}) = e^t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\underline{\Phi}(t; t_0) = \underline{\Phi}(t; 0, \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}) \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} e^t & 4e^{-2t} \\ e^t & e^{-2t} \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} -1 & 4 \\ 1 & -1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -e^t + 4e^{-2t} & 4e^t - 4e^{-2t} \\ -e^t + e^{-2t} & 4e^t - e^{-2t} \end{bmatrix}$$

(ii)  $\alpha_1 = 0$ ,  $\alpha_2$  can be any number

$$\text{if } \underline{\mathbf{x}}_0 \in \text{span} \{ \underline{\mathbf{v}}_2 \} = \text{span} \left\{ \begin{bmatrix} 4 \\ 1 \end{bmatrix} \right\} \Rightarrow \underline{\mathbf{x}}_0 = \alpha \begin{bmatrix} 4 \\ 1 \end{bmatrix} \Rightarrow \underline{\Phi}(t; 0, \alpha \begin{bmatrix} 4 \\ 1 \end{bmatrix}) = \alpha \begin{bmatrix} 4 \\ 1 \end{bmatrix} e^{-2t} \text{ stable!}$$